

WAREHOUSE MANAGEMENT SYSTEM

Build Roadmap

A pipeline-stage snapshot of the WMS build — what's live and running against real data, what's queued, and what's been deliberately parked. This edition also covers a new kind of session: not a new module, but a hardening pass over what's already built, plus the app's first reporting surface.

PIPELINE AT A GLANCE

01  Master Data	02  Yard & Gate	03  Inbound	04  Putaway	05  Inventory	06  Outbound	07  Picking	08  Dispatch	09  Returns	10  Analytics
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Pipeline stages live & verified against real data

6

Stages still queued, incl. Inventory next

1

New reporting surface — Insights, launched this session

3

Real bugs found & fixed in this session's hardening pass

Pipeline Overview

STAGE 01 ● BUILT Master Data Warehouses, SKUs, Customers, Locations, and Users — all five entities, full CRUD, role/warehouse scoping, bulk Excel import, and a live Plan View for generated bin layouts.	STAGE 02 ● BUILT Yard & Gate Management Dock Door + auto-generated staging, Gate In/Out, Vehicle/Driver registration, Yard tracking, detention cost + alerting, seal/signature capture.	STAGE 03 ● BUILT Inbound Order maker, Excel import, and a real ERP push path; vehicle↔order 1:1 mapping; scan-based receiving with Supervisor approve/reject; Complete Inward Process sign-off.	STAGE 04 ● BUILT Putaway ABC/multi-deep-lane-aware bin suggestion, scan-driven staging→bin execution, Multi-SKU Lane Exception workflow — hardened this session against real fragmentation and aging edge cases.	STAGE 05 ● QUEUED Inventory No live on-hand stock view exists anywhere yet — the ledger has real data in it, nothing renders it. Next up in the build order.
STAGE 06 ● QUEUED Outbound Schema exists, no logic/UI. Planned shape: destination + vehicle capacity check (weight & volume), triggering a pick list.	STAGE 07 ● QUEUED Picking Not started.	STAGE 08 ● QUEUED Dispatch Not started.	STAGE 09 ● QUEUED Returns Not started.	STAGE 10 ● QUEUED Analytics Deliberately last, built on top of everything else. Insights — a new, separate reporting page — started ahead of it this session, not a substitute.

This Session: A Hardening Pass

Rather than starting the next pipeline stage, this session ran a different kind of pass — going back through what's already built to find and fix real gaps, one confirmed scenario at a time, each traced through the actual code rather than guessed at.

FIX — PUTAWAY / AGING

Same-day lane top-up, not exact-millisecond

A SKU filling part of a lane one trip, then more of the same SKU arriving on a later trip — even the same day — was silently blocked from continuing that lane, because the aging match defaulted to requiring the exact same millisecond timestamp.

Fixed to same-calendar-day by default. The setting itself moved from **Company** to **Warehouse** once it became clear tolerance genuinely differs per node — and got a real Settings control, the first this field has ever had.

FIX — PUTAWAY / FRAGMENTATION

"Still incoming" lane reservation

A large SKU still mid-delivery could get fragmented across two lanes if a smaller, unrelated SKU happened to arrive in between and got funneled into the same space.

A lane is now reserved to its current occupant while that occupant still has quantity outstanding on an open receipt — reopens automatically the moment delivery completes. Applies to Class B and C.

WORKAROUND — PUTAWAY / CAPACITY

Class B set to single-SKU exclusivity

A related bug was found: a SKU already sharing a capped lane could get wrongly blocked from its own lane's last empty position. Rather than fix the underlying logic immediately, Class B's sharing cap was dropped to match Class A's full exclusivity — sidestepping the bug's trigger condition for now.

The real logic fix stays open, tracked for whenever a class needs a cap above one again.

NEW — REPORTING

Insights launched: Storage Utilization by ABC Class

The app's first dedicated reporting page. For each ABC class, of the lanes currently holding that class's stock, how much of the space is genuinely occupied versus wasted — a direct lens on storage strategy, not just a debug aid.

A new top-level page, not a card bolted onto an existing one — the intended start of a real reporting section.

Built — Master Data & Yard/Gate

Master Data

● BUILT STAGE 01

- **Warehouses** — doubles as a network-node master (Factory/Distributor/Regional DC/National DC/CNF/Cross-dock), storage-type breakdown, dispatch flows, bulk import.
- **SKUs** — category-linked classification, ABC class, multiple storage units, bulk import/export with live summary analytics.
- **Customers** — multi ship-to per customer, per-ship-to GSTIN and delivery-zone tagging.
- **Locations/Bins** — 14-value zone type independent of 5 physical storage builds; a bulk range generator; a live top-down Plan View with real Storage-Type colour-coding and per-flank Rack Name labels.
- **Location Labels** — real, printable Code128 barcodes per bin, closing the loop on what's actually scannable on the floor.
- **Users** — hierarchical role/warehouse-scoped creation and edit, bulk import, an append-only login ledger.

Yard & Gate Management

● BUILT STAGE 02

- **Dock Doors** — fully auto-generated from a warehouse's dock count, including matched Inbound/Outbound staging bins.
- **Gate In/Out** — one combined page, document checks, E-Way Bill, physical condition and seal/signature capture.
- **Vehicle & Driver** — warehouse-scoped registration, blacklist flag, a dedicated Master page for edit.
- **Yard tracking** — live slot occupancy, dock-assignment driver notifications (SMS + voice, stubbed providers).
- **Detention** — company-wide rate with per-vehicle overrides, a real free-hours window, step-function cost, cron-driven alerting proven to fire against real data.

Open: blacklist enforcement, Gate Pass numbering, Dock Scheduling, a Yard Plan View, self-service driver check-in — all schema-ready or scoped, none built.

Built — Inbound & Putaway

Inbound

● BUILT STAGE 03

- **Order creation** — manual maker, Excel bulk import (one file, many orders), and a real per-company API-key ERP push path.
- **Vehicle mapping** — every order requires a vehicle, one open order per vehicle at a time; Match Order auto-finds by vehicle, no typed reference number.
- **Scan-based receiving** — universal capture, tiered interpretation: a clean barcode match auto-accepts; anything else blocks for a Supervisor decision.
- **Reconciliation** — per-line, never a blended total — one short SKU keeps the whole receipt open even if others reconcile perfectly.
- **Complete Inward Process** — a deliberate human sign-off gating Gate Out, distinct from the receipt reaching Received.

Putaway

● BUILT STAGE 04

- **Trigger modes** — Immediate or Batch, company-configurable, per-SKU batch-size override.
- **Bin suggestion** — ABC/multi-deep-lane-aware, deepest-position-first, reservation-aware, prefers the fullest eligible lane, flank-correct on mirrored aisles.
- **Execution** — scan claims a trip, only a matching location scan completes it, no operator override.
- **Multi-SKU Lane Exception** — a real request/approve/revoke workflow bypassing the sharing cap when genuinely needed.
- **This session** — same-day aging by default, per-warehouse aging control, and a "still incoming" lane reservation protecting a large SKU mid-delivery from fragmentation.

Open: Ground/Stillage's own version of this logic, a cancel/exception path, correcting an already-completed mis-putaway, real queue-ordering/aging-based prioritization.

Queued — Not Yet Started

The remaining pipeline stages, in build order. Inventory is next — the ledger already has real data flowing into it from Inbound and Putaway; nothing yet renders it back.

05	Inventory	No live on-hand stock view exists anywhere yet. Even a read-only "what's at Location X" screen would close a real, felt gap — the natural next stage.
06	Outbound	Schema exists, no logic or UI. Planned shape, from an earlier conversation: destination and vehicle-capacity checking (weight and volume), triggering a pick list. Benefits from Inventory existing first, so a "can this order be fulfilled" check means something.
07	Picking	Not started.
08	Dispatch	Not started. Related but distinct: a Dock-out → Gate-out signal has been raised, needing its own design pass before either is built.
09	Returns	Not started.
10	Analytics	Deliberately last, built on top of everything else once there's enough real operational data to analyse. Not the same thing as Insights — a new, separate reporting page launched this session, on purpose, ahead of the full module.

Deferred & Parked

Every item below was deliberately set aside, not forgotten — most need a real design conversation, a missing dependency, or a client decision before they're worth building.

PUTAWAY & STORAGE STRATEGY

- **Self-exclusion cap bug** — the real fix, currently sidestepped by a default change.
- **Per-SKU sharing caps as a client toggle** — completely unwired today; a client can't set it.
- **Ground/Stillage's own multi-position logic** — racked storage came first, on purpose.
- **A cancel/exception path** for a task that can't be completed at all.
- **Correcting an already-completed mis-putaway.**
- **Real queue-ordering/aging-based task prioritization.**

INBOUND

- **Closing a short receipt** — no formal discrepancy close-out path yet.
- **Batch/Lot + expiry capture** — parked until Inventory's own design.
- **Real GS1/unique-barcode parsing** — the tyres/FMCG case-barcode problem.
- **Camera-based scanning** — hardware-scanner input works today; no decode library wired up.
- **erpCode -based resolution** — a small fast-follow once it's wired anywhere.

YARD & GATE

- **Self-service driver check-in** — a top competitor-research gap, "later we do it."
- **Yard Plan View** — needs a spatial-layout pass first.
- **A cancel/void path** for a mistaken Gate In.
- **Dock + Staging + Yard visualizer** — paused mid-conversation, needs real spatial data.
- **Dock Scheduling** — the real dock-selection algorithm; zero design/schema yet.
- **Blacklist enforcement** at Gate In — today's flag is informational only.

PLATFORM-WIDE

- **Real SMS/Email/WhatsApp/Voice providers** — every channel is still a stub that only logs.
- **DockLocationDistance tooling** — the actual dock-suggestion algorithm, deferred until Putaway/Picking need it.
- **Analytics dashboard, WMS/TMS integration** — explicitly parked, not quick wins.
- **SUPER_ADMIN creation, a configurable permission matrix**, real SAP/ERP integration, cloud deployment — all fully deferred.